

Close lenses

X21 (2021) — English translation

The optical system consists of a large number of collecting lenses with a focal length F with a common main optical axis. The centers of adjacent lenses are at the same distance $d \ll F$ from each other, and the thickness of the lens can be neglected compared to d . The ray, before refraction in the center of the first lens, intersects the main optical axis at an angle of $\alpha \ll 1$, and immediately after refraction in the last one, it intersects the main optical axis at the same angle.

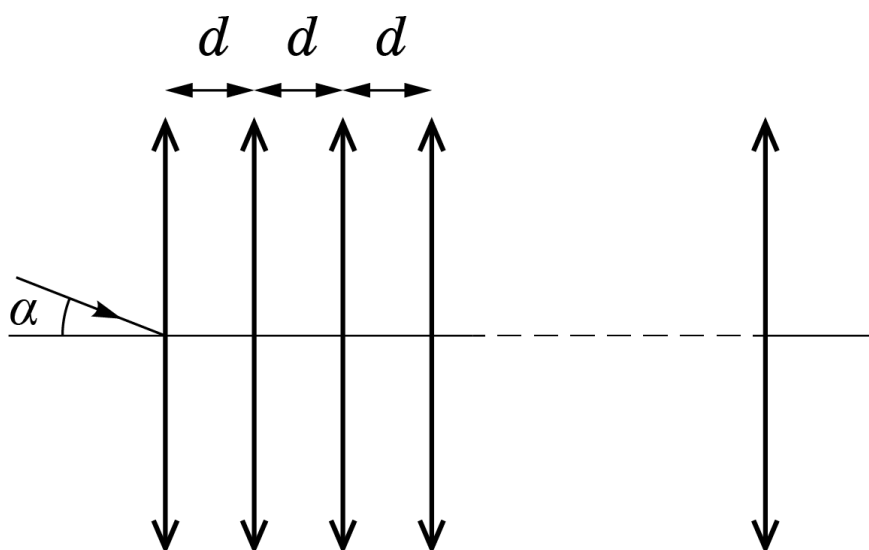


Figure 1: Figure 1.

A1 Find the minimum possible distance L between the outer lenses.
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5.00

Source: <https://pho.rs/p/99>